Fake Product Review Detection using Machine Learning

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***Abstract*—The contribution of fake product reviews in influencing trust and purchasing decisions of consumers on the online marketplace. While recent developments like machine learning and natural language processing have improved the detection of fake reviews, a number of issues such as scalability, multilingual data processing, and ethical implications such as bias and transparency still present important problems in the way of detecting fake reviews. Several previous studies also note features that may not be present like feature diversity, detection in real time, and generalizability across platforms. This study analyzes the existing detection techniques and pinpoints the possibility to develop effective, efficient and globally applicable fake review detection systems.**

Keywords: Fake Reviews, Machine Learning, NLP, Sentiment Analysis, Data Augmentation, Scalability, Ethical AI, E-commerce Fraud Detection.

I. Introduction

This study tackles the escalating problem of counterfeit product reviews and introduces a machine learning framework aimed at detecting misleading content through natural language processing (NLP) methods. It is based on previous research related to data augmentation, data fusion, and classification algorithms like Random Forest. By employing preprocessing techniques such as text cleaning, tokenization, and TF-IDF vectorization, the system converts review texts into features that are amenable to automated evaluation. The objective of this initiative is to offer an effective solution for identifying fraudulent reviews while enhancing trust and transparency within online retail environments. [1] The most recent market analysis also demonstrated that online shopping is one of the fastestgrowing markets worldwide and plays a very important role in the customer perception of products. However, several of these reviews are counterfeit or misleading, with the aim of manipulating product ratings or mislead customers. Its human-like, persuasive quality makes manual detection of such deceptive reviews a struggle. With the advent of the digital marketplace, established moderation methods are insufficient and not sustainable. [2] Machine Learning (ML) has become an effective and reliable technique for detecting fraudulent reviews relying on linguistic features to verify legitimacy. This paper proposes a simple, yet robust machine learning (ML) based Fake Product Review Detection System, which combines Natural Language Processing (NLP), classification techniques including Logistic Regression, SVM, and Random Forest, and thus helps to improve transparency and trust towards online platforms. [3]

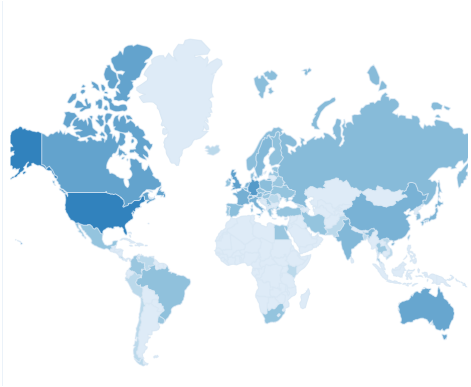

Fig. 1. Most active Countries/Regions

For Target 9.4 in the area of SDG 9, the need to upgrade and modernize industry and digital infrastructure in line with and contributing to sustainability, transparency, and efficiency. User-generated content has played a critical role in online marketplaces of the 21st century, with consumers overwhelmingly using product reviews in their purchases to stay informed and protect platform integrity. Fake review detection based on machine learning is one approach to this target through which it reinforces digital infrastructure, builds data trustworthiness, and encourages innovation in responsible AI systems. E-commerce platforms could help to promote trust, decision making, and a more sustainable digital economy by recognising and minimizing misleading or manipulated reviews. For SDG 12, Target 12.8 focuses on the need to facilitate responsible purchases. False reviews pervert customer perceptions, induce misleading purchasing behavior and wasteful buying behaviors, and encourage unnecessary product returns. One of the targets which we aim to further by developing a detection model is to improve the content accuracy of e-commerce product information that consumers can use to find goods/services more responsible and relevant. The project encourages healthy consumption practices with less misinformation available in digital marketplaces and

encourages ethical online commerce. On SDG 16, Target 16.6 aims to build effective, accountable, and transparent institutions. Digital commerce platforms serve as modern institutions where fairness, accountability, and transparency are crucial to good business practice. Fake reviews undermine these principles by helping dishonest sellers gain an unfair advantage and harming customer trust. With an automatic review-authentication technology, the project achieves institutional integrity, encourages fair competition and minimizes the likelihood for digital manipulation. This directly contributes to Target 16.6 by making online marketplaces more transparent, reliable, and just.

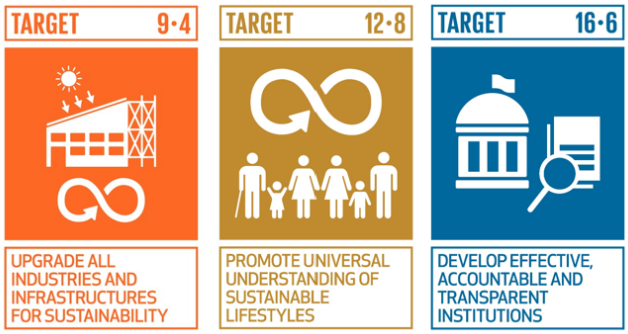

Fig. 2. Goals 9, 12 and 16

II. Literature Review

A. Data Augmentation for Fake Reviews Detection:

Liu and Poesio (2023) carried out a focused investigation on data augmentation to improve fake review detection, especially in cases where available training data is narrow or has limited linguistic diversity. In the world of fake reviews, they often mimic natural writing styles, making detection challenging with very little stylistic diversity for the model. To take care of this problem, the authors employed many augmentation techniques, among them synonym replacement, backtranslation, sentence reordering, random deletion, paraphrasing, and controlled noise injection to generate synthetic examples that maintain the original meaning but differ in structure and vocabulary. [3] These techniques provide a drastic extension of the dataset, as well as giving the model exposure to a wider spectrum of various expressive types to learn more deeply, strong decision boundaries, and generalize unseen data. Their results showed the power of augmentation, decreasing overfitting, and improving accuracy and performance across multiple metrics. It was found that data augmentation is not only beneficial but also core for building scalable and reliable fake review detection systems that can uncover subtle forms of deception that appear on real-world online platforms. [6]

B. Data Fusion for Better Fake Reviews Detection:

Melleng et al. (2023), proposed a data fusion framework that combines textual features with behavioral and metadata signals to improve the detection of fake reviews. They argued that linguistic analysis was not enough to track down deceptive behavior. They use multiple indicators such as reviewer posting frequency, account age, device usage patterns, rating consistency, and unusual activity spikes variables that often expose fraudulent intent regardless of whether the review text is authentic. In conjunction with TF-IDF vectors, sentiment polarity, and stylometric markers, the model learns complex-level patterns that text-only systems typically fail to detect. Their experiments resulted in significant improvements in precision, recall, and reduction of false positives, especially on difficult situations that featured only short and intentionally neutral fake reviews. It was evident that real-world fraud detection practices which are more suited to ecommerce platforms apply to multi-source data fusion, and the fusion could act as a better basis for the next-gen fake review detection systems which need to work on large and diverse populations of users. [4]

C. Fake Review Detection Using Machine Learning:

Rustagi et al. (2022) reviewed different machine learning models to detect fake product reviews like Logistic Regression, Support Vector Machine (SVM), Naïve Bayes, and Random Forest, and determined that having proper preprocessing is crucial to achieve high performance. Their approach includes multiple stages including text cleaning, tokenization, lemmatization, punctuation removal, bigram extraction, and TF-IDF vectorization to transform the unstructured review text into a numerical representation that can be used in machine learning algorithms. Logistic Regression and SVM showed excellent results in handling linear representation of data; however, they were less able to represent complex linguistic data. Random Forest performed better as it can capture non-linear relationships, address noisy user-generated content, and generalize well across review styles. [5] It was also found that hyperparameters such as tree depth and the number of estimators tuning enhanced performance, while ensemble techniques provided a more stable performance on diverse datasets. The authors have shown that machine learning models, especially Random Forest, provide a strong and scalable platform for the detection of deceptive reviews provided that their preprocessing is very accurate and their feature design is good.

D. Security, Interoperability, and Scalability Challenges

Literature consistently points to three major challenges:

- **Security:** Fraudulent reviewers adapt their strategies, requiring continuous updates of detection models.
- **Interoperability:** Lack of standardized datasets across languages and domains limits generalization.
- **Scalability:** Processing millions of reviews across global platforms demands computationally efficient models.

E. Gaps

Despite current advances, it remains a significant challenge to detect fake reviews. Most datasets are limited to the English language and to specific domains (e.g., Yelp, Amazon), thus working inefficiently when considered outside that context. Feature engineering mainly considers text, and behavior or metadata signals are hardly involved, e.g., reviewer history and IP address patterns [4].

TABLE I
SUMMARY OF GAPS IN LITERATURE

Gap	Description
Dataset Limitations	Most datasets are domain-specific (e.g., Yelp, TripAdvisor), with few multilingual or cross-platform corpora.
Feature Engineering	Heavy reliance on textual features; limited integration of behavioral (e.g., reviewer history, IP addresses) or network-based features.
Model Generalization	Models trained on one domain often fail to generalize to others.
Real-time Application	Few studies address the scalability required for real-time detection in commercial platforms.
Ethical Issues	Privacy and cultural sensitivity in data collection remain underexplored.

III. LITERATURE SYNTHESIS

Overall, methodologies based on ML and NLP can identify fake reviews. SVM and Naive Bayes are former methods on which you can build. Meanwhile, accuracy was enhanced by ensemble and deep learning methods Ḃut one technique is not enough: hybrids that blend linguistic, behavioral, and network attributes have the best potential. Future research should prioritize the use of multilingual datasets, the development of scalable real-time systems, and the establishment of ethical frameworks to ensure fairness and transparency. [7]

A. Research Goal

The **goal** is to develop an accurate and scalable detection system that identifies fraudulent reviews, compares model performance, and provides insights to enhance trust and transparency on e-commerce platforms. This study aims to analyze and detect fake product reviews using machine learning. [8] The process involves collecting and preprocessing review data, extracting linguistic and behavioral features, and applying models such as Logistic Regression, SVM, Random Forest, and XGBoost.

IV. PROPOSED SYSTEM: FAKE REVIEW DETECTION

This part describes the setup of the fake review detection system, the end is a trained text classifier prepared for operation.

A. Dataset and Environment

Two experiments were carried out on Google Colab using Python with pandas, NumPy, scikit-learn, matplotlib, seaborn and joblib as main libraries. Amazon product reviews are the type of record which include:

Free-text review. Star rating (1–5). Product category and metadata. A label of the review being authentic or phony.

In the original file, the label column used short codes (CG, OR). These were mapped to more readable labels:

- CG → ”real”
- OR → ”fake”

and then converted to binary variable (label_bin) (real=0, fake=1). Missing-label rows were eliminated to provide a clean training signal.

Shape: (40432, 4)

	category	rating	label	text_
0	Home_and_Kitchen_5	5.0	CG	Love this! Well made, sturdy, and very comfor...
1	Home_and_Kitchen_5	5.0	CG	love it, a great upgrade from the original. I...
2	Home_and_Kitchen_5	5.0	CG	This pillow saved my back. I love the look and...
3	Home_and_Kitchen_5	1.0	CG	Missing information on how to use it, but it i...
4	Home_and_Kitchen_5	5.0	CG	Very nice set. Good quality. We have had the s...

Fig. 3. Label Distribution: Real vs Fake Reviews datasets

B. Data cleaning and preprocessing

Multiple data cleansing steps were implemented toimprove data quality and eliminate noise:

- Duplicate and contradiction handling
- Exact duplicate rows were removed.
- Duplicate review texts (column text) were detected if these appeared with different labels (both real and fake). When comparing texts with different labels, such records of conflicting entries were removed.This reduction of noise reduces label inconsistencies and obviates confusion. [9]

C. Text length filtering

Two length elements were introduced:

- word_len: number of words in the review
- char_len: number of characters

Reviews shorter than 3 words and reviews longer than 500 words were removed, either due to being too short to have any meaning or extreme outliers.

D. Unicode normalization and emoji removal

- The text was normalized using unicodedata.normalize(”NFKD”, ...) and saved in text_norm.
- Emojis and unusual symbols were removed, producing text_no_emoji.

E. Final text normalization

A cleaning function final_clean() was applied to yield a final column clean_text. The procedures include:

- Lowercasing the whole text and removing URLs.
- Removing emojis and punctuation.
- Removing digits.
- Collapsing multiple spaces into a single space.

It yields a normalized text that is tokenizable for bag-of-words features.

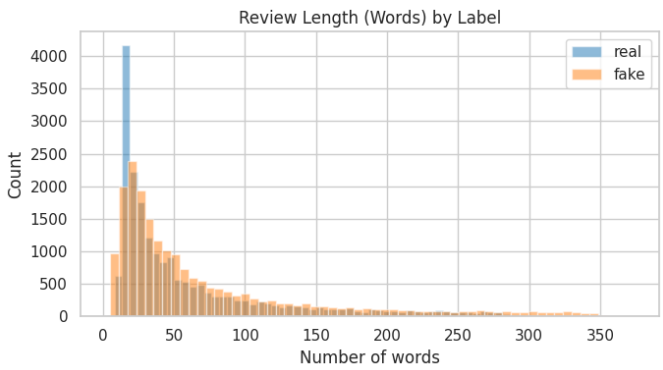

Fig. 4. Histogram of review length by label – use the "Review Length (Words) by Label

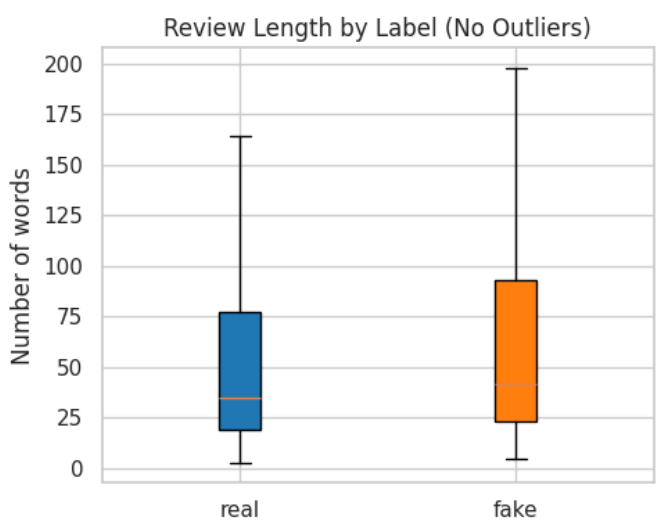

Fig. 5. Boxplot of review length (real vs fake) – shows that fake reviews tend to be slightly longer

F. Exploratory Data Analysis (EDA)

The label balance and the fake review behavior were examined through exploratory analysis:

Label distribution: bar chart displaying counts of real and fake reviews. Rating vs label: grouped bar chart of star ratings (1–5), with real/fake sections to find out if fake reviews are clustered in certain ratings. Length statistics: histograms and boxplots of word_len for each label. Correlation analysis: a correlation heatmap for rating, word_len, char_len, and label_bin shows that review length has a weak positive correlation with the "fake" label, but no single numeric feature is a strong predictor. Category × rating fake rate: a heatmap of fake rate per (category, rating) for the top 6 categories shows which product types and ratings are more affected by suspicious reviews. [10]

1) Feature Extraction with TF–IDF: For text representation, the project uses TF–IDF (Term Frequency–Inverse Document Frequency) via TfidfVectorizer:

- `Input: clean_text column`

  Parameters:
- `max_features = 30000`
- `ngram_range = (1, 2) (unigrams + bigrams)`
- `min_df = 3 (ignore very rare terms)`

  The data was split into 80% training and 20% testing using train_test_split with stratify=y to preserve label proportions. The vectorizer was fit to the training text only and then used to transform both the training and test sets into sparse TF - IDF matrices (X_train_tfidf, X_test_tfidf).

G. Classification Models

Two linear models were trained and compared:

Logistic Regression

- Implemented with `sklearn.linear_model .LogisticRegression`

  Parameters:
- `max_iter = 500`
- class_weight = "balanced" for label imbalance.

  `Outputs: predicted labels (y_pred_lr) and probabilities (y_proba_lr).`

Support Vector Machine (SVM) Implemented with `sklearn.svm.LinearSVC` It was trained on the same TF–IDF features. Outputs are: predicted labels (y_pred_svm).

Both models were tested on held-out test set by:

- Accuracy.
- Precision.
- Recall.
- F1-score.

Hyperparameter	Value
max_features	30000
ngram_range	(1, 2)
min_df	3

TABLE II
TF–IDF VECTORIZER HYPERPARAMETERS.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9352	0.9255	0.9466	0.9359
SVM	0.9448	0.9403	0.9500	0.9451

TABLE III
PERFORMANCE COMPARISON OF LOGISTIC REGRESSION AND SVM ON THE TEST SET.

SVM obtained higher scores for all metrics and was chosen as the final model.

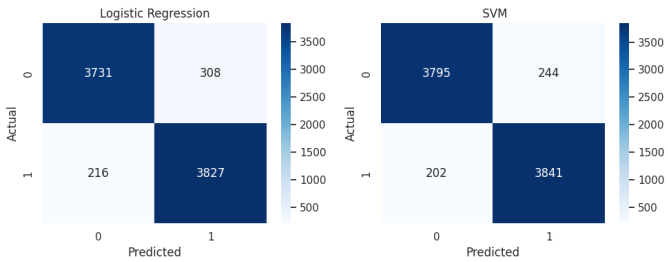

Fig. 6. confusion matrix - Logistic Regression

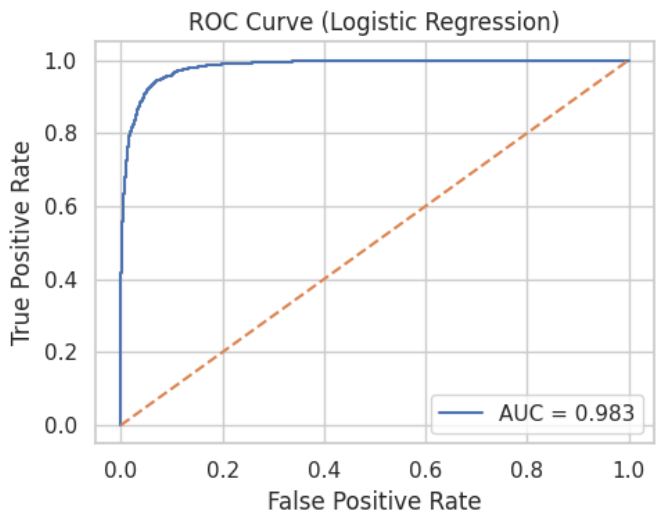

Fig. 7. ROC Curve -Logisitc Regression

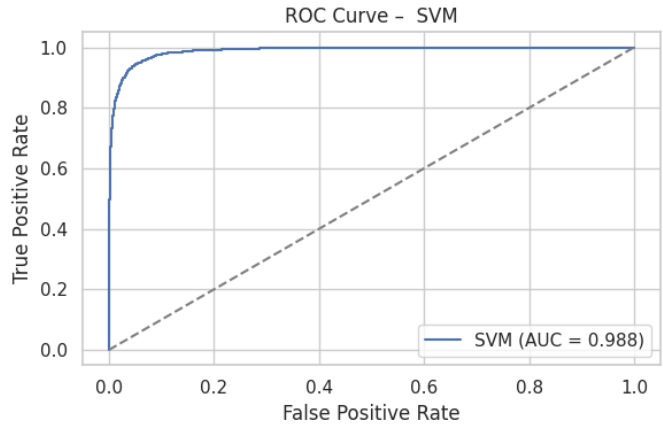

Fig. 8. ROC Curve -SVM

V. ERROR ANALYSIS AND MODEL SAVING

To better understand model behavior, a small sample of test reviews was inspected:

- A DataFrame test_df_sample displays random test reviews along with their true label, predicted label, and probability of being fake.

This helps identify cases where the model is confident but wrong, and which types of reviews are difficult to classify.

Finally, the selected Linear SVM model and the TF–IDF vectorizer were saved using joblib:

- final_model.pkl – trained classifier
- tfidf_vectorizer.pkl – fitted TF–IDF transformer

VI. WEB APPLICATION FOR FAKE REVIEW DETECTION

To apply the model for the practical side, a web-based interface was created. In the system users can enter a product review and immediately get a prediction if the review is real or fake. Here, we showcase the machine learning model in real time and provide a simple, user-friendly experience to demonstrate the practical utility of our fake review detection platform.

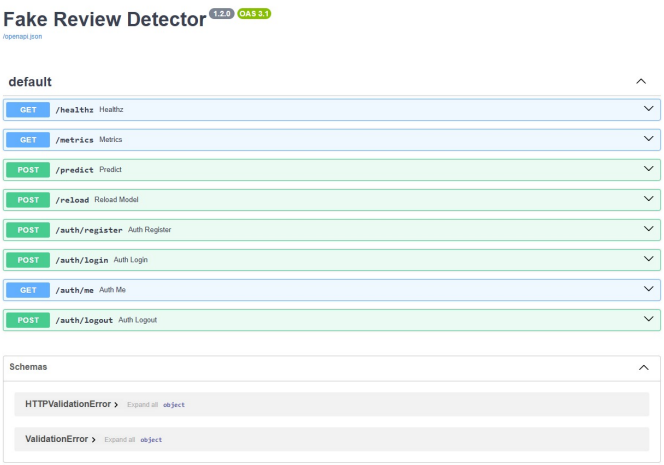

Fig. 9. API

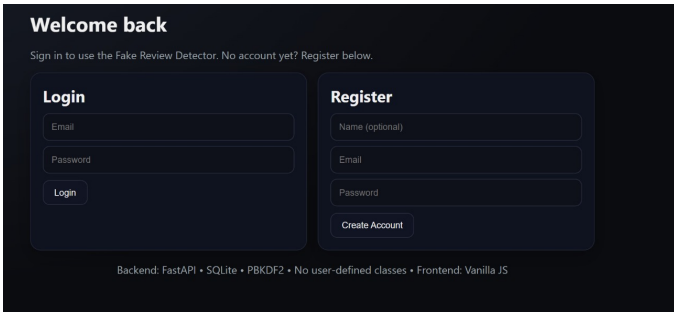

Fig. 10. Login

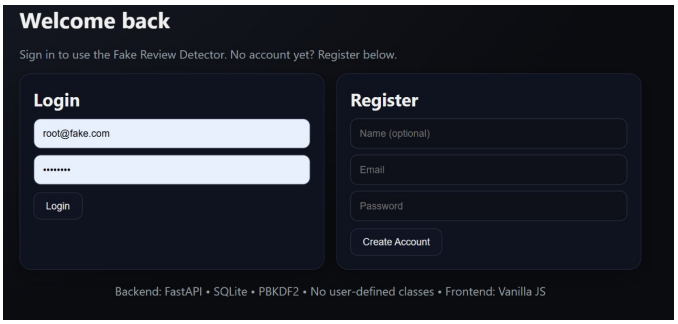

Fig. 11. Login page

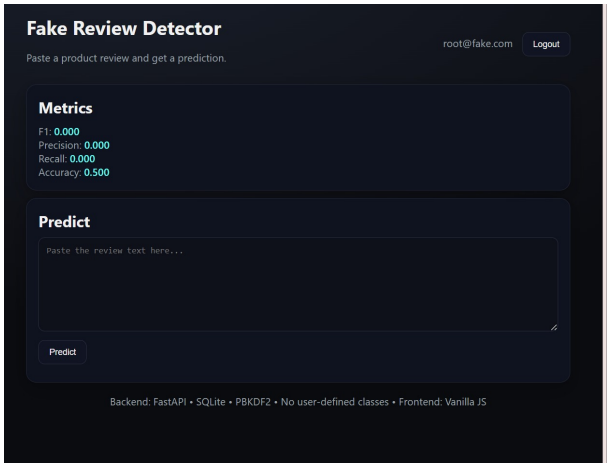

Fig. 12. main page

Fig. 13. Testing (real review)

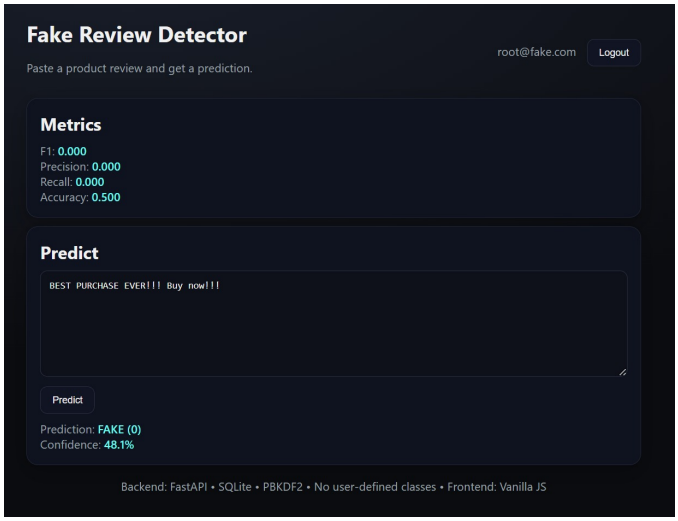

Fig. 14. Testing (fake review)

VII. Conclusion

The reviewed studies highlight the growing importance of developing reliable methods for detecting fake product reviews, particularly as e-commerce platforms continue to expand. Prior research demonstrated that improving dataset quality through data augmentation, integrating multiple data sources through data fusion, and applying robust machine learning models such as Random Forest can significantly enhance classification accuracy. Together, these findings show that both linguistic and behavioral features play a crucial role in identifying deceptive content effectively. [11] Based on these insights, our project builds on previous work by implementing a machine learning–based Fake Product Review Detection System that uses NLP preprocessing and classification algorithms to distinguish between genuine and fake reviews. By adopting techniques supported by existing research, the system aims to improve transparency, reduce misinformation, and strengthen trust within online shopping platforms. Ultimately, this contributes to a safer and more credible digital marketplace for consumers and businesses alike. [12]

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